Assignment_8 递归

完成学生：庞又玮 物理学院

题目

`M04147`：汉诺塔问题(Tower of Hanoi)

dfs, http://cs101.openjudge.cn/pctbook/M04147

递归的模板题...

思路：

1. 把上面 n−1 个盘从 A 移到 B（借助 C）；
2. 把最大的第 n 个盘从 A 移到 C；
3. 把 n−1 个盘从 B 移到 C（借助 A）.

```
def hanoi(n: int, src: str, aux: str, dst: str, moves: list) -> None:
    if n == 0:
        return

    hanoi(n - 1, src, dst, aux, moves)  # 1
    moves.append(f"{n}:{src}->{dst}")  # 2
    hanoi(n - 1, aux, src, dst, moves)  # 3


num, a, b, c = input().split()
num = int(num)
lst = []
hanoi(num, a, b, c, lst)
print("\n".join(lst))
```

代码运行截图 （至少包含有"Accepted"）



M05585：晶矿的个数

matrices, dfs similar, http://cs101.openjudge.cn/pctbook/M05585

```
def dfs(x: int, y: int, ch: str):
    if x < 0 or x >= n or y < 0 or y >= n:
        return
    if m[x][y] != ch or visited[x][y]:
        return
    visited[x][y] = True  # 标记已遍历过的元素（套路）
    for dx, dy in [(1, 0), (-1, 0), (0, 1), (0, -1)]:
        dfs(x + dx, y + dy, ch)


k = int(input())
for _ in range(k):
    n = int(input())
    m = [list(input()) for _ in range(n)]
    ans_1 = ans_2 = 0
    visited = [[False] * n for _ in range(n)]
    for i in range(n):
        for j in range(n):
            if not visited[i][j] and m[i][j] == 'r':
                ans_1 += 1
                dfs(i, j, 'r')
            if not visited[i][j] and m[i][j] == 'b':
                ans_2 += 1
```

```
                dfs(i, j, 'b')
    print(ans_1, ans_2)
```

代码运行截图 （至少包含有"Accepted"）



#53632585提交状态

查看 提交 统计 提问

状态: Accepted

源代码

```
def dfs(x: int, y: int, ch: str):
    if x < 0 or x >= n or y < 0 or y >= n:
        return
    if m[x][y] != ch or visited[x][y]:
        return
    visited[x][y] = True
    for dx, dy in [(1, 0), (-1, 0), (0, 1), (0, -1)]:
        dfs(x + dx, y + dy, ch)


k = int(input())
for _ in range(k):
    n = int(input())
    m = [list(input()) for _ in range(n)]
    ans_1 = ans_2 = 0
    visited = [[False] * n for _ in range(n)]
    for i in range(n):
        for j in range(n):
            if not visited[i][j] and m[i][j] == 'r':
                ans_1 += 1
                dfs(i, j, 'r')
            if not visited[i][j] and m[i][j] == 'b':
                ans_2 += 1
                dfs(i, j, 'b')
    print(ans_1, ans_2)
```

基本信息

#: 53632585
题目: M05585
提交人: 26n2600011560
内存: 3612kB
时间: 29ms
语言: Python3
提交时间: 2026-10-27 20:23:32

M02786：Pell数列

dfs, dp, http://cs101.openjudge.cn/pctbook/M02786/

```
MOD = 32767
n = int(input())
queries = [int(input()) for _ in range(n)]
max_k = max(queries)

# 递归易爆栈，换用迭代
pell = [0] * (max_k + 1)
if max_k >= 1:
    pell[1] = 1
if max_k >= 2:
    pell[2] = 2
for i in range(3, max_k + 1):
    pell[i] = (2 * pell[i - 1] + pell[i - 2]) % MOD

ans = [pell[k] for k in queries]
```

```
for num in ans:
    print(num)
```

代码运行截图 （至少包含有"Accepted"）



#53633342提交状态

查看 提交 统计 提问

状态: Accepted

源代码

```
MOD = 32767
n = int(input())
queries = [int(input()) for _ in range(n)]
max_k = max(queries)

# 递归易爆栈，换用迭代
pell = [0] * (max_k + 1)
if max_k >= 1:
    pell[1] = 1
if max_k >= 2:
    pell[2] = 2
for i in range(3, max_k + 1):
    pell[i] = (2 * pell[i - 1] + pell[i - 2]) % MOD

ans = [pell[k] for k in queries]
for num in ans:
    print(num)
```

基本信息

#: 53633342
题目: M02786
提交人: 26n2600011560
内存: 40476kB
时间: 306ms
语言: Python3
提交时间: 2026-10-27 21:07:05

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English 帮助 关于

M46：全排列

backtracking, https://leetcode.cn/problems/permutations/

回溯的模板题...

```
class Solution:
    def permute(self, nums: list[int]) -> list[list[int]]:
        n = len(nums)
        ans, sol = [], []

        def backtrack():
            if len(sol) == n:
                ans.append(sol[:])
                return

            for x in nums:
                if x not in sol:
                    sol.append(x)  # 选择
                    backtrack()  # 递归
                    sol.pop()  # 撤销选择（回溯）
```

```
        backtrack()
        return ans
```

or

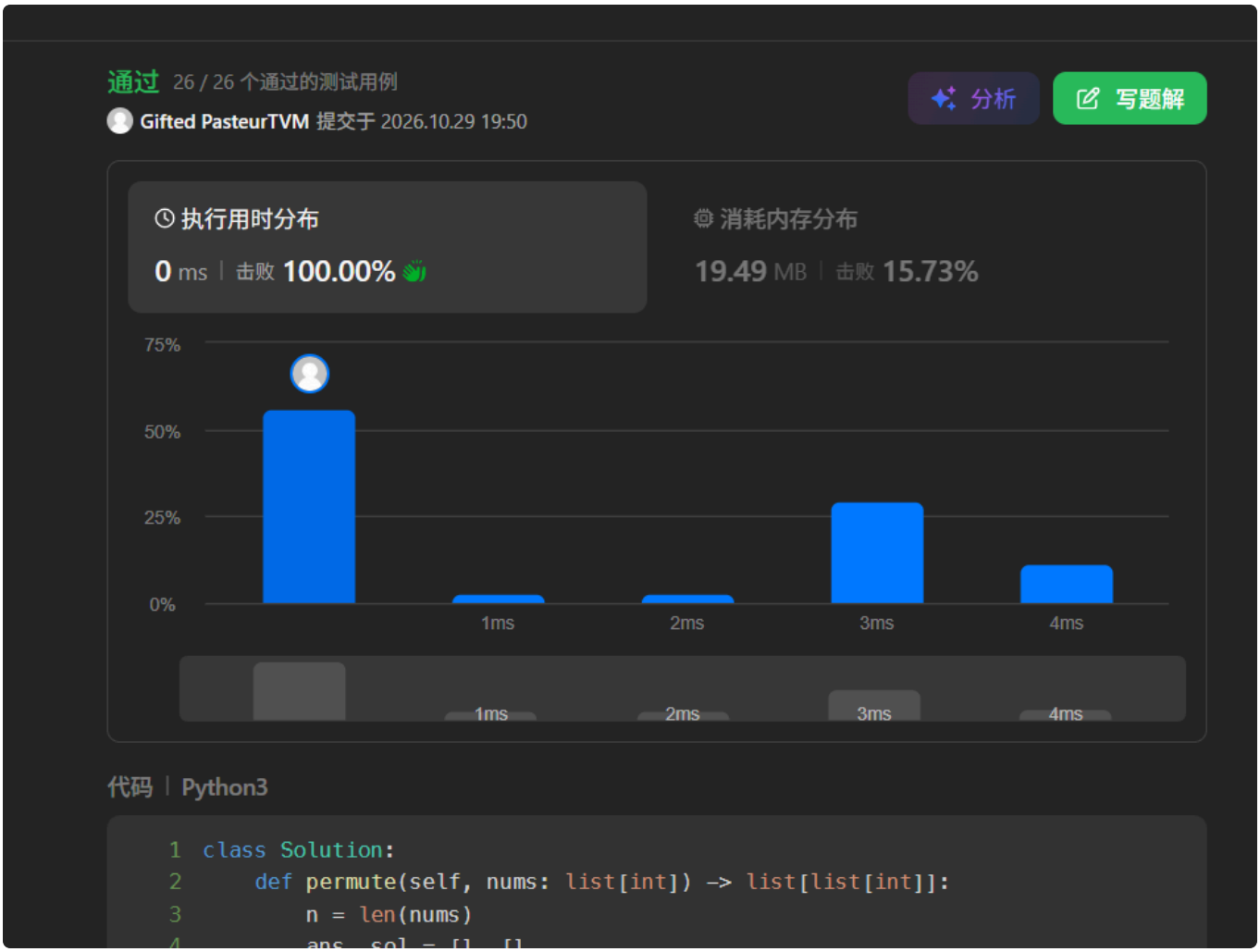
```
class Solution:
    def permute(self, nums: list[int]) -> list[list[int]]:
        n = len(nums)
        ans, sol = [], []
        used = [False] * n

        def backtrack():
            if len(sol) == n:
                ans.append(sol[:])
                return

            for i in range(n):
                if not used[i]:
                    used[i] = True
                    sol.append(nums[i])  # 选择
                    backtrack()  # 递归
                    sol.pop()  # 撤销选择（回溯）
                    used[i] = False

        backtrack()
        return ans
```

代码运行截图 （至少包含有"Accepted"）



T02754：八皇后

dfs and similar, http://cs101.openjudge.cn/pctbook/T02754

```
cols = [0] * 8
placement = []


def dfs(row):
    if row == 8:
        placement.append(int("".join(str(c + 1) for c in cols)))
        return

    for col in range(8):
        for i in range(row):
            if cols[i] == col or abs(cols[i] - col) == row - i:
                break
        else:
            cols[row] = col
            dfs(row + 1)
            cols[row] = 0
```

```
dfs(0)
n = int(input())
for _ in range(n):
    k = int(input())
    print(placement[k - 1])
```

在DeepSeek老师的引导下写对的~

代码运行截图 （至少包含有"Accepted"）



T01958 : Strange Towers of Hanoi

http://cs101.openjudge.cn/practice/01958/

```
def strange_hanoi(n: int, a: str, b: str, c: str, d: str):
    if n == 0:
        return 0

    moves = []
    for k in range(1, n + 1):
```

```
        moves.append(strange_hanoi(n - k, a, d, c, b) + (2 ** k - 1) +
strange_hanoi(n - k, b, a, c, d))
    return min(moves)

for i in range(1, 13):
    print(strange_hanoi(i, "A", "B", "C", "D"))
```

谁懂菜菜的我做好几个小时最后终于成功独立做出tough题的救赎感呀！

代码运行截图 （至少包含有"Accepted"）

